SWEEP rule on the KD-Toric code

April 10, 2026

1 Notations.

We follow the standard notations commonly used in the literature. For integer n, we denote by $[n]$ the set of integers $\{1, 2, \ldots, n\}$; for integer i, we denote by 1_i the vector consisting of one at the ith coordinate and zero elsewhere.

When not otherwise mentioned, we consider quantum states over qubits, each lying in a two-dimensional Hilbert space $\mathcal{H}_2$, and denote by $\mathcal{H}_{2^n}$ the Hilbert space of an n-qubit system. Pure quantum states and their dual presentations are denoted by $|\psi\rangle, \langle\psi|$, and in the matrix representation, we use $|\psi\rangle\langle\psi|$ to denote its density matrix. We will use ρ to denote a mixed state, whose density matrix is a trace one PSD matrix.

Below, a semi proof that in CSS code partial correction $Z^e \to Z^{\hat{e}}$ doesn't add bit flip errors. It might adds a global phase. I should enter it in a section about CSS coding.

$$\begin{aligned}
& X^f Z^e \left|w + C_z^\perp\right\rangle \\
& = X^f Z^e X^w \left|C_z^\perp\right\rangle \\
\text{apply } H^{\otimes n} & \mapsto Z^f X^e Z^w \left|C_z\right\rangle \\
& = (-1)^{e\cdot w} Z^f Z^w X^e \left|C_z\right\rangle \\
\text{partial correction } X^{e+\hat{e}} & \mapsto (-1)^{(e+\hat{e})(f+w)} (-1)^{e\cdot w} Z^f Z^w X^{\hat{e}} \left|C_z\right\rangle \\
\text{apply } H^{\otimes n} & \mapsto (-1)^{(e+\hat{e})(f+w)} (-1)^{e\cdot w} X^f X^w Z^{\hat{e}} \left|C_z^\perp\right\rangle \\
& = (-1)^{(e+\hat{e})(f+w)} (-1)^{e\cdot w} (-1)^{\hat{e}\cdot w} X^f Z^{\hat{e}} X^w \left|C_z^\perp\right\rangle \\
& = (-1)^{(e+\hat{e})f} X^f Z^{\hat{e}} \left|w + C_z^\perp\right\rangle
\end{aligned}$$

2 Computation Model

Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, in different space-time locations. Each space-time location, associated with a gate, encodes the time of the application and the qubits in the support of the gate. Formally

Separate between an element gate $g \in \mathcal{G}$ and a gate, which also has a space-time location.

Definition 2.1 (Quantum Circuit)**.** A space-time location l is a tuple that encodes in its first coordinate time, and in its second coordinate a subset of integers referring to a support of (q)bits. For a collection of quantum gates $\mathcal{G} = \{g \colon L(\mathcal{H}_2^{\otimes\Delta}) \to L(\mathcal{H}_2^{\otimes\Delta})\}$, and integers $w, d \in \mathbb{N}$ a quantum circuit C, with depth d and width w, is a collection of tuples $\{u_i = (g_i, l_i)\}$, where u_i are the gates of C and each u_i is composed of a gate $g_i \in \mathcal{G}$ and a space-time location $l_i \in \mathbb{Z}_d \times \mathbb{Z}_w^\Delta$. The gates of C have to satisfy the constraint that for every two gates with the same time coordinate, there cannot be a non-empty intersection in their space coordinate, namely for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies $l_{i,2} \cap l_{j,2} = \emptyset$. We define the t-cut of C to be $C|_t := \{u_i = (g_i, l_i) \colon u_i \in C \text{ and } l_{i,1} = t\}$.

To define the running of a quantum circuit C on a given (mixed) quantum state, we will need to have the notion of the acyclic directed graph associated with C.

Definition 2.2 (Associated Computation DAG)**.** Let $C = \{u_i\}$ be a quantum circuit with depth d and width w. We define the associated computation DAG of C, denoted by π_C, to be the acyclic directed graph $\pi_C = (\mathcal{U}, E)$ as follows:

1. The vertex set $\mathcal{U}$ is the gate set of C, namely $\mathcal{U} = \{u_i\}$.
2. For $u_i = (g_i, l_i), u_j = (g_j, l_j)$, There is an edge $\{u_i, u_j\} \in E$ if and only if:

$$j \in \arg\min_j \{l_{j,1} : l_{j,1} > l_{i,1} \text{ and } l_{j,2} \cap l_{i,2} \neq \emptyset\}$$

   Note that there might be multiple indices that get the minimum, so the out degree might be greater than one.

In addition, suppose that $J = \{j_1, j_2, \ldots, j_k\} \subset [|C|]$ is an order of the indices of the gates in C according to the topological order induced by π_C. Then, we say that J is an order of indices that agrees with π_C and define the series of circuit $C_J = \{C_{J,i}\}$ as follows:

1. The first element: $C_{J,1} = \{u_{j_1}\}$
2. Then, $C_{J,i+1} = C_{J,i} \cup \{u_{j_i}\}$. We name $C_{J,i}$ the partial computation circuit until step i.

Definition 2.3. Let ρ be a density matrix over w qubits, and let $C = \{u_i\}$ be a quantum circuit with depth d and width w. For any density matrix σ, quantum gate $g \in \mathcal{G}$, and location l, the application of the gate-location tuple (g, l) on σ, denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ, denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}$ on ρ, according to a topological order induced by π_C.

We say that $\rho_1, \rho_2, \ldots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u_i' \circ \rho_i$, where $\{u_i'\}$ is a prefix of a topological ordering of $\{u_i\}$. If $\{u_i'\}$ contains all the gates in $\{u_i\}$, then we call $\rho_1, \rho_2, \ldots, \rho_k$ a running.

Definition 2.4 (C Subjected to $\mathcal{E}$)**.** Let $C = \{u_i\}$ be a quantum circuit with depth d and width w. For a set of fault locations $\mathcal{E} \subset [d] \times [w]$, we define the quantum circuit C *subjected to* $\mathcal{E}$, denoted by $C[\mathcal{E}] = \{u'_i\}$, to be the quantum circuit that is the result of the following map:

$$u_i = (g_i, (l_{i,1}, l_{i,2})) \mapsto \begin{cases} (g_i, (2l_{i,1}, l_{i,2})) & u_i \in C \\ (g_i, (2l_{i,1}-1, l_{i,2})) & u_i \in \mathcal{E} \end{cases}$$

We will denote the associated computation DAG of C affected by $\mathcal{E}$ as $\pi[C, \mathcal{E}]$.

Definition 2.5 (Faults Propagation to a Step τ)**.** For a quantum circuit $C = \{u_i\}$, with depth d, an order $J = \{j_1, j_2, \ldots, j_k\} \subset [|C|]$ that agrees with π_C, and gate u_i in C, a propagation of a gate $u_i \in C$ to step τ is the quantum operator $X\colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$ obtained by the follow process:

1. First, check if i precedes τ in J. If not, then return $X = I$. Otherwise, denote $j_l = i$, and look for an operator X_1 such that $u_{j_l} u_{j_{l+1}} = u_{j_{l+1}} X_1$.
2. Then, for $k \leq \tau - l$, look for an operator $X_k\colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$, such that $X_{k-1} u_{j_{l+1}} = u_{j_{l+1}} X_k$.
3. At the final stage of the process, namely where $k = \tau - l$, we define $X := X_k$ to be the propagation of u_j to step τ.

For a quantum circuit C, a set of faults $\mathcal{E} = \{e_i\}$, an order $J \subset [|C[\mathcal{E}]|]$ that agrees with $\pi_{C[\mathcal{E}]}$, and a step $\tau \in J$, denote by $X_{e_i}^\tau$ the propagation of $e_i \in \mathcal{E}$ in $C[\mathcal{E}]$ to step τ. Let $j'_1, j'_2, \ldots, j'_l$ be the indices of the faults in $\mathcal{E}$ such that their corresponding positions in J precede τ, ordered according to the reverse topological order induced by $\pi_{\mathcal{E}}$. The propagated error to step τ is defined by

$$X^\tau := \prod_i X_{e_{j'_i}}^\tau$$

Definition 2.6 (Deviation)**.** For a quantum circuit C, a set of faults $\mathcal{E}$, and step τ, let X^τ be the propagated error at step τ. We define the deviation of $C[\mathcal{E}]$ from C at step τ by the set of indices on which X^τ acts non-trivially.

Definition 2.7. Using the same notations as in Definition 2.6, for a set of unitary operators $\mathcal{S} = \{s\colon \mathcal{H}_2^w \to \mathcal{H}_2^w\}$, we define the propagated error up to stabilizers in $\mathcal{S}$ as $X^\tau[S] := \min_{s \in \mathcal{S}}\{w(sX)\}$. Then, we define the deviation of $C[\mathcal{E}]$ from C at step τ up to stabilizers in $\mathcal{S}$ to be the subset of coordinates on which $X^\tau[S]$ acts non-trivially.

Definition 2.8 (Generalized Tanner Graph)**.** For registers a and b, their generalized Tanner graph is their 'union' $\mathcal{T}_a \cup \mathcal{T}_b$. This allows us to define the Tanner graph of a quantum circuit C, at step τ, by stacking together Tanner graphs. This also allows us to define the Tanner graph if a register. rewrite. add definition for a register R of quantum circuit C.

Definition 2.9 (Clustered Deviation)**.** For a quantum circuit C, a register R of C, and a set of faults $\mathcal{E}$, let $\mathcal{T}_R$ be the Tanner graph of R, and let $\mathcal{D}_\tau$ be the deviation of $C[\mathcal{E}]$ from C on R at step τ. Denote by $\mathcal{T}_R|_{\mathcal{D}_\tau}$ the subgraph of $\mathcal{T}_R$ induced by the restriction of the left vertices (bits) to the indices in $\mathcal{D}_\tau$. Let us abuse and extend the connection notion to Tanner graphs by defining that two left vertices are connected if and only if they share a right neighbor. We define the deviation clusters at step τ to be the connected components of $\mathcal{T}_R|_{\mathcal{D}_\tau}$. The size of each deviation cluster is the number of right vertices it contains.

Claim 2.10. Let C be a quantum circuit composed of gates with bounded fan degree $\Delta \in \mathbb{N}$. In addition, let R be a quantum register encoded by a quantum error correction code $\mathcal{C}$ with Tanner graph $\mathcal{T}$ that has left and right degrees Δ_1 and Δ_2. Fix a set of faults $\mathcal{E}$, and an order J that agrees with $\pi_{C[\mathcal{E}]}$. For any τ, denote by $Y_1^{(\tau)}, Y_2^{(\tau)}, \ldots, Y_l^{(\tau)}$ the deviation clusters at step $\tau \in J$. Assume that at step $\tau_1 \in J$ there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}|$. Then there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z, such $Z \cap Y_i^{(\tau_1)} \neq \emptyset$ and Z is of size at least:

$$|Z| \geq \frac{1}{\Delta\Delta_1\Delta_2}|Y_i^{(\tau_1)}|$$

Furthermore there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z', such $Z' \cap Y_i^{(\tau_1)} \neq \emptyset$, and Z' is of size at most:

$$|Z'| \leq \Delta\Delta_1\Delta_2|Y_i^{(\tau_1)}|$$

Add: that when it isn't mentioned, we assume that the error at time t can be decomposed into a product $\Rightarrow$ at each step, a fault is supported on at most one (q)bit.

Proof. Denote by $u = (g, l)$ the gate at step τ. If u acts trivially on the (q)bits in $Y_i^{(\tau_1)}$ then they also belong to deviation on register R at step $\tau_1 - 1$, Since they are connected in $\mathcal{T}$ they form a deviation cluster on register R at step $\tau_1 - 1$ and we got the desired. Thus it is left to handle the case in which u acts non-trivaily on (q)bits in $Y_i^{(\tau_1)}$. Since u acts on at most Δ qubits, and each qubit has at most $\Delta_1(\Delta_2 - 1)$ neighbors at $\mathcal{T}$ it follows that u acts non trivialy on at most

$$\Delta + \Delta\Delta_1(\Delta_2 - 1) \leq \Delta\Delta_1\Delta_2$$

deviation clusters at step $\tau_1 - 1$. Let l be integer no greater than $\Delta\Delta_1\Delta_2$ and let $Y_1, Y_2, \ldots, Y_l$ be the deviation cluster, on register R, at step $\tau_1 - 1$, on which u acts non trivialiy, since $Y_i^{(\tau_1)} \subset \bigcup_j Y_j$. We have that:

$$\begin{aligned} |Y_i^{(\tau_1)}| &\leq \sum_j |Y_j| \leq \Delta\Delta_1\Delta_2 \max_j |Y_j| \\ \Rightarrow \max_j |Y_j| &\geq \frac{1}{\Delta\Delta_1\Delta_2}|Y_i^{(\tau_1)}| \end{aligned} \tag{1}$$

Complete the second inequality. Should consider to erase what have already written about it. To get the second inequality, we use the fact that u is a unitary gate, in particular, u is reversible. Let $C_{J,\tau_1} = u_1, .., u_{\tau_1}$ be the partial computation circuit until step τ_1. Observe that $C_{J,\tau_1 - 1}$ is obtained by stacking $u^{\dagger}_{\tau_1}$ as a final stage to C_{J,τ_1}, denoted by $u^{\dagger} \circ C_{J,\tau_1}$. Thus, the deviation from C_{J,τ_1} at step $\tau_1 - 1$ equals the deviation from $u^{\dagger} \circ C_{J,\tau_1}$ at step $\tau_1 + 1$. By eq. (1), $\square$

Corollary 2.11. Let $\alpha = 1/(\Delta\Delta_1\Delta_2)$. Assume that at the initial step j_1 there is no deviation cluster of size more than αd, and that at a step $\tau_1 \in J$ there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}| > d$. Then there is a step $\tau_o < \tau_1, \tau_o \in J$ for which there is a deviation cluster Z at step τ_o, such that the size of Z is in the range $(\alpha d, d)$.

Proof. $\square$

Claim 2.12. Let A be the event in which, until time t, there is a time t' such that there is a deviation cluster of size greater than d. For any τ, let B_τ be the event that step τ is the first step to has a deviation cluster of size in the range $(\alpha d, d)$. Then:

$$\mathbf{Pr}\left[A\right] \leq 2wd \max_{\tau \in [2wd]} \mathbf{Pr}\left[B_\tau\right]$$

Proof.

$$\mathbf{Pr}\left[A\right] \leq \mathbf{Pr}\left[\bigcup_{\tau} B_\tau\right] \leq 2wd \max_{\tau \in [2wd]} \mathbf{Pr}\left[B_\tau\right]$$

$\square$

Change letters: the d in the model section refers to a size that is comparable to the code distance where here d refers to the dimension of the Toric complex

3 The (d, d')-Dimensional Toric Code and the SWEEP Rule

Definition 3.1 (d-Dimensional Toric Complex)**.** For integers d and n, denote the group resulting from the d-directed power of the cyclic group of order n by $\mathcal{G}_n^d = \mathbb{Z}_n^{\times d}$, and denote the standard generating set of $\mathcal{G}_n^d$ by $S = \{1_i \colon i \in [d]\}$. We define the *$d$-dimensional toric complex* to be the hypergraph obtained from the Cayley graph $\mathrm{Cay}(\mathcal{G}_n^d, S)$ by taking its vertices as the 0-faces and for any $i \in [d-1]$ we define the i-faces as follow: First, for a vertex $v \in \mathcal{G}_n^d$ and for a subset of generators $S' \subset S$ of size $|S'| = i$, the i-face rooted at v and spanned by S', denoted by $\theta_{v,S'}$, is:

$$\theta_{v,S'} := \bigcup_{I \subset S'} \left\{ \left(\prod_{g \in I} g \right) v \right\}$$

The i-faces of the complex are the collection of all the possible rooted spanned i-faces induced by $\mathrm{Cay}(\mathcal{G}_n^d)$. We will denote them by $\mathcal{F}_i$.

Additionally, we name n and d, the *side length* and the *dimension* of the complex.

Definition 3.2 (Positive Edges and Cubes)**.** For $u, v \in \mathcal{G}_n^d$, and $S_1', S_2' \subset [d]$, with sizes $|S_1'| = |S_2'| = i$, we say that the i-faces $\theta_{v,S_1'}$ and $\theta_{u,S_2'}$ are adjacent if the intersection $\theta_{v,S_1'} \cap \theta_{u,S_2'}$ is an $i-1$-face. In addition, if θ_1 and θ_2 are i and $i-1$-faces such that $\theta_2 \subset \theta_1$, then we say that θ_2 is a *positive edge relative to face* θ_1 if they are rooted at the same vertex. Furthermore, we say that θ_1 is a *positive cube* of θ_2. When the i-face is clear from the context, we will abbreviate and say that θ_2 is *positive*.

Example 3.3. For example, let $v, u \in \mathcal{G}_n^d$ and $g_1 \neq g_2 \in S$. Consider the 2-face $\theta = \{v, g_1 v, g_2 g_1 v, g_2 v\}$. Then, for $x, y \in \mathcal{G}_n^d$, we say that an edge (1-face) $\{x, y\}$ is *positive relative to the face* θ if $x = v$ and y is either $g_1 v$ or $g_2 v$.

We denote the complex by $G = (V, E, F)$, where V, E, and F are the 0, 1, and 2 faces.

Add a paragraph that explains that we choose not to use a chain-complex since we earn nothing from it i.e don't care about topology properties. Yet mention that chain-complex formulation proved itself as an effective tool to compute the properties of the codes we use.

Definition 3.4 (Assignment on The (d', d)-Toric Complex)**.** Let $\mathcal{T}$ be a d-dimensional complex. Each of its i-face sets naturally induces a vector space $\mathbb{F}_2^{|\mathcal{F}_i|}$. We call a binary vector $z \in \mathbb{F}_2^{|\mathcal{F}_i|}$ an assignment on the i-faces of $\mathcal{T}$. For a vertex v and a subset of generators $S' \subset S$ of size $|S'| = i$, we use the notation $\theta_{\mathbf{v},\mathbf{S'}}$ to represent the unit vector in $\mathbb{F}_2^{|\mathcal{F}_i|}$ that has 0 in every coordinate except the one associated with the i-face $\theta_{v,S'}$. Then

$$\left\{\theta_{\mathbf{v},\mathbf{S'}} : v \in \mathcal{F}_0, S' \subset S \text{ s.t } |S^{'}| = i\right\}$$

is a base to $\mathbb{F}_2^{|\mathcal{F}_i|}$ Not exactly base, they are linear dependent. Let z be an assignment on the i-faces. Let $v_1, \ldots, v_k \in \mathcal{F}_0$ and $S_1, \ldots, S_k \subset S^{\times k}$ be the generator subsets such that $\theta_{v_1,S_1'}, \theta_{v_2,S_2'}, \ldots, \theta_{v_k,S_k'}$ are the collection of i-faces that support z, namely:

$$z = \sum_{i=1}^{k} \theta_{\mathbf{v_i},\mathbf{S_i'}}$$

We refer to $\theta_{v_1,S_1'}, \theta_{v_2,S_2'}, \ldots, \theta_{v_k,S_k'}$ as the collection of faces that form z, or briefly, the faces form z. For $i > 0$, we define the linear map $\partial : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i-1}|}$ such that

$$\partial\theta_{\mathbf{v},\mathbf{S'}} := \sum_{g \in S'} \theta_{\mathbf{v},\mathbf{S'/g}} + \sum_{g \in S'} \theta_{\mathbf{gv},\mathbf{S'/g}}$$

For $i < d$, we define the linear map $\partial^* : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i+1}|}$ such that

$$\partial^*\theta_{\mathbf{v},\mathbf{S'}} := \sum_{g \in S/S'} \theta_{\mathbf{v},\mathbf{S' \cup g}} + \sum_{g \in S'} \theta_{\mathbf{g^{-1}v},\mathbf{S' \cup g}}$$

We define the *restriction of z to a vertex v* to be the linear map $|_v \colon \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_i|}$ such that:

$$\theta_{\mathbf{u},\mathbf{S'}}|_v := \begin{cases} \theta_{\mathbf{u},\mathbf{S'}} & v \in \theta_{u,S'} \\ 0 & \text{else} \end{cases}$$

We define the *front of z to a vertex v* to be the linear map $|_{v+} \colon \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_i|}$ such that:

$$\theta_{\mathbf{u},\mathbf{S'}}|_{v+} := \begin{cases} \theta_{\mathbf{u},\mathbf{S'}} & u = v \\ 0 & \text{else} \end{cases}$$

We say that an assignment z is *connected* if the graph that is induced by taking the faces that support z as vertices and defining the edges to be the adjacency relation between them is connected. We define the connected components of z to be its subsets such that each of them is a connected assignment. For a j-face f, we say that $f \in z$ if z has a support on $\theta_{\mathbf{v},\mathbf{S'}}$ such that $f \in \theta_{v,S'}$. Consider an assignment $z \in \mathbb{F}_2^{\mathcal{F}_i}$ and a vertex $v \in \mathcal{G}_n^d$. We say that z is a *rooted assignment* at vertex v if there are subsets $S'_1, \ldots, S'_k \subset S^{\times k}$ such that each of S'_j is at size i and z is decomposed as:

$$z = \sum_{j}^{k} \theta_{\mathbf{v},\mathbf{S'_j}}$$

Put differently, z is the result of taking a collection of i-faces that are rooted at v.

Add a proof for $\partial\theta_{\mathbf{v},\mathbf{S'}} \cdot \partial^*\theta_{\mathbf{u},\mathbf{S''}} = 0 \Rightarrow$ gives a CSS code.

Need to decide on what should I elaborate, for example should I provide a proof for the distance and the rate of the code? [Den+02]

Definition 3.5 ((d, d') Toric Code)**.** Let d, d' be integers such that $1 < d' < d$. For any integer n, we construct the nth instance of the quantum code family (d, d') *Toric Code* as follows: Let $G = (\mathcal{F}_0, \ldots, \mathcal{F}_d)$ be the d-dimensional toric complex obtained from $\mathcal{G}_n^d$ as defined in Definition 3.1. We associate the (q)bits with the $\mathcal{F}_{d'}$, the X-checks with the $\mathcal{F}_{d'-1}$, and the Z-checks with the $\mathcal{F}_{d'+1}$. An X-check, c_x, is satisfied if the parity of (q)bits set on the d'-face containing c_x is even. Similarly, a Z-check, c_z, is satisfied if the parity, in the Hadamard basis, of the (q)bits set on the d'-faces contained in c_z is even.

Definition 3.6 (Sweep Rule)**.** Let z be an assignment over the d'-faces, The *sweep rule decoding procedure* at vertex $v \in \mathcal{G}_n^d$, defining as follow:

1. For the $\mathcal{C}_X$ code. Look for a rooted assignment $\tilde{z}$ at v such that $(\partial\tilde{z})|_{v+} = \partial(z)|_{v+}$. Namely, the X-checks rooted at v get the same syndromes for z and $\tilde{z}$. If there is one, then flip $\tilde{z}$.

2. Apply the Hadamard gate on any d' face containing v.

3. For the $\mathcal{C}_Z$ code. Look for a rooted assignment $\tilde{z}$ at v such that $(\partial^* \tilde{z})|_{v+} = \partial^*(z)|_{v+}$. Namely, the Z-checks rooted at v get the same syndromes for z and $\tilde{z}$. If there is one, then flip $\tilde{z}$.

4. Apply the Hadamard gate again on any d' face containing v to reverse the second stage.

Stages 1 and 3 involve measurement of stabilizers.

Definition 3.7 (Decoding Gadget)**.** Consider a quantum circuit C, for a quantum gate u in C, with fun-in Δ, we say that u is a *decoding gadget* if it has the following form:

1. The wires of u wires can be decomposed into two subsets A and B such that the wires in A are reset wires, and there exists an encoded register R in C such that the wires in B are all set in R.

2. For a set of faults $\mathcal{E}$, there is a Pauli gate u' that acts only on B such that the action of u in $C[\mathcal{E}]$ equals u'. Put differently, the quantum circuit obtained from $C[\mathcal{E}]$ by replacing u with u' equals $C[\mathcal{E}]$.

Definition 3.8 (Unitary Decoding Representation)**.** Let $C = \{u_i\}$ be a quantum circuit, and let $\mathcal{U} = \{u_i'\} \subset C$ be a subset of the gates in C that are decoding gadgets. For a set of faults $\mathcal{E}$, we define the *unitary decoding representation* of $C[\mathcal{E}]$ to be the quantum circuit obtained by replacing each of the gates in $\mathcal{U}$ with their corresponding Pauli gate according to $\mathcal{E}$.

Claim 3.9. The local gates performing Toom's rule are decoding gadgets.

Proof. prove that. □

4 Infinite Torics

Definition 4.1 (Registers Coordinates System)**.** Let $C = \{u_i\}$ be a quantum circuit, with width w and registers $\mathcal{R} = R_1, \ldots, R_k$, each of length at most n. For register R_i, we define the register identifier $\mathbf{1}_{R_i} = \mathbf{1}_i$. Then, we define the map $\phi_{\mathcal{R}} \colon \mathbb{Z}_w \to \mathbb{Z}_R \times \mathbb{Z}_n$ to return, for a space-location coordinate $x \in \mathbb{Z}_w$, the tuple consisting of the identifier of the register containing x, and its relative coordinate within the register. Namely,

$$\psi_{\mathcal{R}}(x) = (\mathbf{1}_{R_j}, x')$$

Where R_j is the register containing x, and x' is the index of the qubit, set at the x location, in R_j. The presentation of C in the *registers coordinate system* is the encoding by a set of gates $\{u_j'\}$, which is the result of replacing each space-location of the original gates with the applications of $\psi_{\mathcal{R}}$ on them. Namely, for each $u_i = (g_i, l_i) \in C$, we map its space-location coordinate $l_{i_2} = (l_{i_2}^{(1)}, \ldots, l_{i_2}^{(\Delta)})$ to:

$$l_{i_2} \mapsto (\psi_{\mathcal{R}}(l_{i_2}^{(1)}), .., \psi_{\mathcal{R}}(l_{i_2}^{(\Delta)}))$$

For brevity, we name the first and second coordinates in the result of ψ_R the *register coordinate* and the *inner coordinate*.

Definition 4.2 (Shifting in Registers System)**.** Consider the register coordinate system induced by a quantum circuit of width w, with R registers, each of length at most n. For an integer $r \in \mathbb{Z}$, we define the r-shifting function $\mathrm{Sh}_r \colon (\mathbb{Z}_R \times \mathbb{Z}_n)^{\times \Delta} \to (\mathbb{Z}_R \times \mathbb{Z})^{\times \Delta}$ to act on space-location coordinates as follows:

$$\mathrm{Sh}_r(l_1, .., l_k) = (\mathrm{Sh}_r(l_1), .., \mathrm{Sh}_r(l_k))$$
$$\text{Where } \mathrm{Sh}_r(l_i) = \begin{cases} l_i + r & \text{if } l_i \text{ is an inner coordinate} \\ l_i & \text{else} \end{cases}$$

Definition 4.3. Let n be an integer, and let $C = \{u_i\}$ be a quantum circuit with registers $\mathcal{R} = R_1, \ldots, R_k$ such that each register R_i is a toric encoding with a side length n. In addition, all the decoding gadgets of C are sweep rule gadgets; let us denote them by $\mathcal{U}$. We define the ∞*-extension of* C, denoted by C^∞, as follows:

1. For a register $R_i \in \mathcal{R}$, denote by d_i, d'_i the dimension of the toric complex that induces the code, and the dimension of the faces that are associated with the bits. Each register R_i in $\mathcal{R}$ is mapped to a register R'_i, which is a (d_i, d'_i)-dimensional, infinite side length, toric encoding. Since the mapping between registers is one-to-one, we use the identifier $\mathbf{1}_{R_i}$ to identify also R'_i.

2. Each gate $u_i \in C/\mathcal{U}$ is mapped to:

$$u_i = (g_i, l_i) \mapsto \bigcup_{j=-\infty}^{\infty} \{(g_i, (\mathrm{Sh}_{jn}(l_i))\}$$

3. Each gate $u \in \mathcal{U}$ is mapped to the collection of sweep rule gadgets $\{v_j\}_{-\infty}^{\infty}$ such that each of them has the same time coordinate as u, is in the register that corresponds to the image of u's register, and is rooted at $\mathrm{root}(v) + jn$.

For a set of faults $\mathcal{E} = \{f_i\}$, we denote by $C^\infty[\mathcal{E}^\infty]$ the ∞*-extension of* $C[\mathcal{E}]$.

Claim 4.4. Let $u = (g, l)$ and $v_0, v_{\pm 1}$ plantation of u at locations $\mathrm{root}(u)$ and $\mathrm{root(u)} \pm n$. Then $v_0 v_{\pm 1}|_{[w]} = u$.

Proof. Prove it ☐

Definition 4.5. For a quantum circuit C, of the form defined in Definition 4.3, and a set of faults $\mathcal{E}$, denote:

$$\rho := C[\mathcal{E}] \left|0\right\rangle\!\left\langle 0\right| C[\mathcal{E}]^\dagger$$
$$\rho_\infty := C^\infty[\mathcal{E}^\infty] \left|0\right\rangle\!\left\langle 0\right| (C^\infty[\mathcal{E}^\infty])^\dagger$$

We say that a quantum circuit C is ∞*-compatible* if, for any set of faults $\mathcal{E} = \{f_i\}$, it holds that:

$$\rho = \mathbf{Tr}_{/[w]} \left(\rho_\infty\right)$$

Claim 4.6. Let C be quantum circuit of the form defined in Definition 4.3. Then C is ∞-compatible.

Proof. Its enough to prove the claim for circuits in their unitary decoding representation. We prove it by induction on the number of the gates.

1. Base. For an empty circuit, ρ_∞ is just the infinite string of zeros; thus, the restriction of it to $[w]$ locations, namely $\mathbf{Tr}_{/[w]}(\rho_\infty)$, yields a single matrix element corresponds to the w-length string of zeros, which is exactly ρ.

2. Step. Assume the correction of the claim for any circuit, with less than τ steps, and consider a circuit C with τ steps. Denote by C' the circuit obtained form C by erasing the last gate in C, observes that C' is a circuit with $\tau - 1$ steps. Denote by ρ' and ρ'_∞ the: Enter it. By the induction assumption we have:

$$\rho' = \mathbf{Tr}_{/[w]}\left(\rho'_\infty\right)$$

Denote the last gate in C by u, and by $I = \{v_j\}_{-\infty}^{\infty}$ the collection of gates in C^∞ to which u is mapped. Let us split into cases, depending on whether u is or is not a decoding gate.

(a) Suppose that u is not a decoding gate. Let g and l be the gate element and the location of u. By the definition of ∞-extension, I equals:

$$I = \bigcup_{j=-\infty}^{\infty} \{(g, (\mathrm{Sh}_{jn}(l))\}$$

So there is only one gate in I, corresponding to $j = 0$, that has support on qubits with spatial location in the range $[w]$. Denote that gate by v_0.

$$\begin{aligned}
\mathbf{Tr}_{/[w]}\left(\rho_\infty\right) &= \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty}^{\infty} v_j \rho'_\infty \prod_{j=-\infty}^{\infty} v_j^\dagger\right) \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty, j\neq 0}^{\infty} v_j \rho'_\infty \prod_{j=-\infty, j\neq 0}^{\infty} v_j^\dagger\right) v_0^\dagger \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty, j\neq 0}^{\infty} v_j^\dagger \prod_{j=-\infty, j\neq 0}^{\infty} v_j \rho'_\infty\right) v_0^\dagger \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\rho'_\infty\right) v_0^\dagger \\
&= v_0 \rho' v_0^\dagger \\
&= \rho
\end{aligned}$$

(b) In the case that u is a decoding gate, observe that there are at most two gates in I that have support on the qubits with space-location in $[w]$. If there is exactly one, then repeating the non-decoding gate case while setting this gate as v_0 gives the desired result. Else, in the case where two gates act non-trivially on qubits at $[w]$, denote those gates by v_0, v_1. Since we assumed that C is given in its unitary decoding representation, it holds that v_0, v_1 are Paulis, in particular a product operator. Denote their product decompositions by $v_0 = \prod_k v_0^{(k)}$ and $v_1 = \prod_k v_1^{(k)}$. Thus, by repeating the non-decoding case, but extracting from the infinite product only the terms in v_0 and v_1 that act non-trivially at $[w]$, we get:

$$\mathbf{Tr}_{/[w]}\left(\rho_\infty\right) = v_0 v_1|_{[w]} \rho' \left(v_0 v_1|_{[w]}\right)^\dagger$$

Moreover, since v_0 and v_1 are a plantation of g in locations $\mathrm{root}(u)$ and $\mathrm{root}(u) \pm n$, then by Claim 4.4, we have that $u = v_0 v_1|_{[w]}$; therefore:

$$\begin{aligned}\mathbf{Tr}_{/[w]}\left(\rho_\infty\right) &= v_0 v_1|_{[w]} \rho' \left(v_0 v_1|_{[w]}\right)^\dagger \\ &= u \rho' u^\dagger \\ &= \rho\end{aligned}$$

□

add that the computation graph is lifted naturally to the ∞-extension, since the image of a mapping are overlapped gates, we can execute them simultaneously.

5 Shrinking

consider changing the name to something like hight lines / contour in order to hint on the similarity to topography map. In addition, the embedding inside $\mathbb{R}^d$ should appears before that definition.

Definition 5.1 (Potential Walls)**.** We define the potential walls to be a functions collection from the vertices of $\mathcal{G}_n^d$ to the reals as follow:

1. For any coordinate $i \in [d]$, we define $L_i \colon \mathcal{G}_n^d \to \mathbb{R}$ to act as $L_i(v) = v_i$.
2. For any subset of coordinates $I \subset [d]$ we define $L_I \colon \mathcal{G}_n^d \to \mathbb{R}$ to act as $L_I(v) = -\sum_{i \in I} v_i$.

For a subset of vertices $\Gamma \subset \mathcal{G}_n^d$, a vertex $v \in \Gamma$, and a potential wall $l \colon \mathcal{G}_n^d \to \mathbb{R}$, we say that v is a *local maximum* of l in Γ if for any neighbor u of v, such that $u \in \Gamma$, we have $l(v) \geq l(u)$. We say that v is a strong local maximum if the inequality is strict, namely $l(v) > l(u)$. When the subset Γ is clear from the context, we might omit it and briefly say that v is a local maximum of l.

What we want is that if a bit contains both v, gv then there will be a check that contains both v, gv, that is true if $d' > 1$, then it means that we can earse from the bit a vertex which it is not gv.

Claim 5.2. Let z be an assignment, and let ∂ be the set of vertices adjacent to unsatisfied checks induced by z. Let v be a vertex in ∂, and let g be a coordinate in S such that v is a local maximum of L_g. Then $gv \notin \partial$.

Proof. Suppose $gv \in \partial$, then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to v being a local maximum of L_g in ∂. □

Claim 5.3. Let z be an assignment, and let ∂ be the set of vertices adjacent to unsatisfied checks induced by z. Let v be a vertex in ∂, and let g be a coordinate in S such that v is a local maximum of L_g. Then for any check associated with a $d' - 1$ face rooted at v that contains gv, it is satisfied.

Proof. Assume, through contradiction, the existence of an unsatisfied check associated with a $d' - 1$ positive face rooted at v, which contains gv. This implies $gv \in \partial$, contradicting Claim 5.2. □

Definition 5.4 (Small Code at Vertex v)**.** For a vertex v, we define the *small code at vertex* v to be the binary assignments over the bits (d' faces) rooted at v that satisfy the $\mathcal{C}_X$'s restrictions rooted at v ($d' - 1$ faces). We denote by $\mathcal{C}_v$ the small code at v.

Definition 5.5. We define Stab_v to be the stabilizers rooted at vertex v, namely

$$\mathrm{Stab}_v := \mathrm{Span}\left(\left\{z : z = \partial\theta_{\mathbf{v},\mathbf{S'}}, S^{prime} \subset S, |S^{prime}| = d' + 1\right\}\right)$$

Claim 5.6. Let v be a vertex. For any $z \in \mathcal{C}_v$, there is a stabilizer $w \in \mathrm{Stab}_v$ such that $z = w|_v$; namely, z and w agree on the faces rooted at v.

Proof. For $d = 4, d' = 2$. For $i, j \in [d]$, such that $i \neq j$, denote the bit rooted at v and spanned by generators $g_i, g_j \in S$, namely $\theta_{v,\{g_i,g_j\}}$, by $g_{ij} = g_{ji}$. The restrictions:

$$r_i \colon \sum_{j \neq i} g_{ij}$$

Thus we get

$$\begin{aligned}\sum_{i}^{n} r_i &= \sum_{i}^{n} \sum_{j \neq i}^{n+1} g_{ij} \\ &= \sum_{i}^{n} \sum_{j<i}^{n} g_{i,j} + g_{j,i} + \sum_{i}^{n} g_{i,n+1} \\ &= r_{n+1}\end{aligned}$$

Thus the dimension of the small code at v is at least:

$$\binom{d}{2} - (d-1) = 3$$

We have $\binom{4}{3} = 4$ stabilizers rooted at v (each for each cube). Yet:

$$\begin{aligned} s_1 &= g_{1,2} + g_{2,3} + g_{1,3} \\ s_2 &= g_{1,2} + g_{2,4} + g_{1,4} \\ s_3 &= g_{1,3} + g_{3,4} + g_{1,4} \\ s_4 &= g_{2,3} + g_{3,4} + g_{2,4} \end{aligned}$$

And clearly

$$s_4 = s_1 + s_2 + s_3$$

So the dimension of the stabilizer is 3, and for any stabilizer w rooted at v, we have that $w|_v$ is a codeword of the small code at v. Thus, we can extend z_v to w such that $z_v = w_v$. Since v is a local maximum of L_{d+1}, it follows that $z|_v$ is rooted at v. Therefore, $(z + w)|_v = 0$, namely $v \notin z + w$. Moreover, any $u \in w/w_v$ belongs to $\theta_{gv,J}$, thus $L_{d+1}(u) < L_{d+1}(v)$. □

Claim 5.7. Let z be a finite assignment. Suppose that $\max_{v \in z} L_{d+1}(v) > \max_{v \in \partial z} L_{d+1}(v)$, Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that $\max_{v \in \tilde{z}} L_{d+1} < \max_{v \in z} L_{d+1} - 1$.

Proof. Let $v_1, v_2, \dots, v_k$ be the vertices in z that achieve the maximum for L_{d+1}, namely $v_1, v_2, \dots, v_k = \arg\max_{v \in z} L_{d+1}(v)$, and assume that $v_i \notin \partial z$ for any $i \in [k]$. Consider the ith vertex v_i, by definition, since $v_i \notin \partial z$, we have $(\partial z|_{v_i})|_{v_i} = 0$, and since v_i is a maximum point of L_{d+1} in z, it is also a local maximum, and therefore $z|_{v_i+} = z|_{v_i}$. Combining the two, we get that $z|_{v_i}$ is a codeword of the small code at v, namely $z|_{v_i} \in \mathcal{C}_{v_i}$. On one hand, for any stabilizer w rooted at v, we have $(\partial w)|_{v_i} = 0$; particularly, $(\partial w)|_{v_i+} = 0$, and therefore $w|_{v_i} \in \mathcal{C}_v$. On the other hand, $\dim \mathcal{C}_{v_i} = \dim \mathrm{Stab}_{v_i}$. Then, since $z|_{v_i} \in \mathcal{C}_{v_i}$, it follows that there must be a stabilizer $w_i \in \mathrm{Stab}_{v_i}$ such that w_i and z agree on the positive faces of v_i, namely $w_i|_{v_i} = z|_{v_i}$. That is wrong..

Set $\tilde{z} \leftarrow z + \sum_i w_i$. Since $\sum_i w_i$ is a stabilizer, we have $\tilde{z} \sim_{C_X} z$. In addition, since for any $j \neq i$, $v_j \notin w_i$ we have $\tilde{z}|_{v_i} = (z + w_i)|_{v_i} = 0$ namely $v_i \notin \tilde{z}$, thus:

$$\max_{u \in \tilde{z}} L_{d+1}(u) \leq \max_{u \in z \cup_i w_i / \cup v_i} L_{d+1}(u) < \max_{u \in z} L_{d+1}(u) - 1$$

□

Claim 5.8. Let z be a finite assignment. Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that $\max_{v \in \tilde{z}} L_{d+1} = \max_{v \in \partial \tilde{z}} L_{d+1}$.

Proof. Apply Claim 5.7 multiple times. □

Claim 5.9. Let z and ξ be the assignment and the result of the application of the Sweep rule on it. Then:

1. For any $i \in [d]$, $\max_{v \in \partial z} L_i(v) = \max_{v \in \partial \xi} L_i(v)$.

2. $\max_{v\in\partial z} L_{d+1}(v) > \max_{v\in\partial\xi} L_{d+1}(v) + 1$.

Proof. First, since the Sweep rule is invariant to stabilizers, for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, the correction applied by the Sweep rule on $\tilde{z}$ is the same as the application on z. Thus, $\partial\tilde{\xi} = \partial\xi$, and we can assume by Claim 5.8 that $\max_{v\in z} L_{d+1} = \max_{v\in\partial z} L_{d+1}$.

Thus, the local view around the edge v of ∂z is rooted at v, therefore, v can flip by that and local view and get $v \notin z \to v \notin \partial z$, and eventually we repeat on the same max inequality above to show that L_{d+1} decreases. ☐

Proof.

$$L_{d+1}(u) \le \min_{v\in\partial z} L_{d+1}(v)$$

Observes that if the *termination-condition* holds then any local maximum of L_{d+1} in ∂z is also a local maximum of L_{d+1} in $\tilde{z}$ and we finish. Consider the other case, namely there is at least one vertex $u \in \tilde{z}/\partial z$ such that u is a local maximum for L_{d+1} and

$$L_{d+1}(u) > \min_{v\in\partial z} L_{d+1}(v)$$

In that case, we peak such u, and looking for stabilizer w such $u \notin \tilde{z} + w$ by repeating over the above procedure where u plays the role of v. We keep repeating on the procedure until the *termination-condition* holds.

We argue that the process indeed terminates. Consider a vertex u such that u is a local maximum of L_{d+1} in $\tilde{z}$ and u is not a local maximum of L_{d+1} at z. Put differently, u becomes a local maximum as a result of adding w to z. Thus, u has to belong to $w|_v \setminus v$, which implies that there is a non-empty subset of generators $I \subset S$ such that $u = (\prod_{g\in I} g)v$, and therefore $L_{d+1}(u) \le -1 + L_{d+1}(v)$.

Hence, since z is finite.

We repeat the process while possible; namely, we stop when all the local maxima of L_{d+1} are in ∂. ☐

We prove that $\dim \mathcal{C}_v = \dim \mathrm{Stab}_v$ below, for now assume it is correct. We continue by using the fact that if, for a local maximum v in ∂, it holds that $z|_v$ is rooted at v, then the decoder will find bits rooted at v such that their ∂ agrees with ∂.

To generalize, one might think that it has to complete the local view on v such that all the checks are satisfied. For C_X, it sums up to show that the following dimension is positive. Yet the trick fails for C_Z, where we need to think of a different idea.

$$d\binom{d-1}{d'} - \binom{d}{d'-1} - d\binom{d-1}{d'-1}$$

Claim 5.10. Let z an assignment rooted at vertex v, such $\partial z|_{v+} = 0$. Then there is a stabilizer w, rooted at v, such that $w|_{v+} = z$.

Proof. Denote by F the faces in z, and by J the collection of generators that support F. We claim that $|J| \ge d' + 1$.

Clearly, $|J| \geq d'$, since each face is defined by a root vertex and d' generators. Yet, observe that if $|J|$ equals exactly d', then z contains a single d'-face, denote it by f. Then, any check associated with a $d' - 1$ face of f is unsatisfied, in particular the $d' - 1$ faces that are rooted at v, in contradiction to $\partial z|_{v+} = 0$.

Now, one can choose $J' \subset J$ of size $d' + 1$, and define the stabilizer $w = \theta_{v,J'}$. □

Claim 5.11. Suppose $v \in z$ and for any $g \in S$, we have $g^{-1}v \notin z$, then there is $\theta_{v,J} \in \partial z$.

Proof.

$$\partial z = \sum_{S'} \theta_{\mathbf{v},\mathbf{S}'} + ..$$

□

Claim 5.12. Let z be a connected assignment on the d'-faces at finite size, and let Γ be the set of vertices which are adjacent to unsatisfied checks induced by z. Let v be a vertex in $\mathcal{G}_\infty^d$ such that v is a strong global maximum of L_{d+1}, at Γ. Then there is a rooted assignment $\tilde{z}$ at v such that $z_v = \tilde{z}_v$.

Proof. Let $\Theta = \theta_{v_1,S'_1}, \theta_{v_2,S'_2}, ..\theta_{v_k,S'_k}$ be the faces forms z.

Since z is finite, it has a maximum for L_{d+1}. Denote by u the vertex that gets the maximum, and denote by $\theta_{u,J_1}, \ldots, \theta_{u,J_l}$ the d'-faces in z that are rooted at u. We claim that $u = v$.

Suppose not, namely $u \notin \Gamma$. Then it means that any $d' - 1$ face containing u is satisfied. Observe that $\sum \theta_{u,J_i} \neq 0$; otherwise, it would contradict the non linearity independence of $\theta_{\mathbf{v_1},\mathbf{S'_1}}, \theta_{\mathbf{v_2},\mathbf{S'_2}}, ..\theta_{\mathbf{v_k},\mathbf{S'_k}}$. So, by Claim 5.11 there exists a $d' - 1$ face rooted at u, contained in $\sum \theta_{u,J_i}$. Let us denote it by $\theta_{u,J'}$.

Since $\theta_{u,J'}$ is satisfied, it has to be adjacent to a turned bit, namely a d' face in z, not in $\theta_{u,J_1}, \ldots, \theta_{u,J_l}$. Denote that d' face by f. Yet, since $\theta_{u,J'}$ is rooted at u, it contains u, and therefore f contains u. Now, if f is rooted at u, then it is a contradiction that $\theta_{u,J_1}, \ldots, \theta_{u,J_l}$ are all the d' faces in z rooted at u. If f is not rooted at u, denote its root by w, so there is $g \in S$ such that $gw = u$, and $w \in z$, in contradiction to u being maximum for L_{d+1}.

So, we find that v is also the maximum of L_{d+1} in z. Therefore, we have that $\tilde{z} = \sum \theta_{u,J_i}$ zeros out z_v. □

Claim 5.13. Let z be a connected assignment on the d'-faces at finite size, and let the collection $\Theta = \theta_{v_1,S'_1}, \theta_{v_2,S'_2}, ..\theta_{v_k,S'_k}$ be the decomposition forms z. Denote by Γ the set of vertices adjacent to unsatisfied check induced by z. Then Θ is acyclic, if and only if z has at least one Infimum.

Proof. First, suppose that z has at least one Infimum. Let u be an infimum in z.. Second, suppose that Θ is acyclic. $\square$

Claim 5.14. Suppose that $d' > 1$ and a vertex $v \in \mathcal{G}^d_\infty$ is also adjacent to the domain wall, namely $v \in \Gamma$. It holds that v cannot be simultaneously both a (strong) local max of L_{d+1} and a local min of L_{d+1}.

Proof. Let $v \in \Gamma$ be a local maximum point of L_{d+1}, It means that:

1. The vertex v is adjacent to a d'^{-1}-face which is unsatisfied.

2. That face is rooted in v; otherwise, the root of the face is a vertex in Γ with a higher value of L_{d+1}, in contradiction to v being a local maximum.

Thus, there is a subset of generators $S' \subset S$, of size $d'-1 > 0$, such that the unsatisfied check set is on the face $\theta_{v,S'}$. Given $d' > 1$, we have that S' contains at least one more element besides the empty set, denoted by g. Now, consider the vertex gv. Since gv is adjacent to $\theta_{v,S'}$, then $gv \in \Gamma$. On the other hand, on the infinite toric $\mathcal{G}^d_\infty$, moving along the grid generator decreases L_{d+1} by one, so $L_{d+1}(gv) = L_{d+1}(v) - 1$. In other words, gv is a neighbor of v with a lower value of L_{d+1}, and thus v cannot be a local minimum of L_{d+1}. $\square$

Remark 5.15. That claim alone gives a decoder in the clean setting. Write a remark indicating that when $d' = 1$, meaning qubits are on the edges (2D toric), the claim is not true.

Definition 5.16 (Reduced subset with boundrey at v)**.**

6 Space-Time Graph

Definition 6.1. Consider a graph $G = (V, A, F)$, where V is a set of vertices, and A, F are sets of undirected edges. For $v \in V$, denote by $\mathrm{pre}(v) = \{u \in V : \{u, v\} \in A\}$, we extend the notion for subsets of vertices $X \subset V$ by $\mathrm{pre}(X) = \bigcup_{v \in X} \mathrm{pre}(v)$. Let $T : V \to \mathbb{N}$, We say that (G, T) is a time graph if and only if:

1. $\{x, y\} \in A \Rightarrow T(y) = T(x) \pm 1$

2. For $x, y \in V$ $\{x, y\} \in F$ if and only if there exists $z \in V$ such both $\{x, z\}, \{y, z\} \in A$.

We will call to T time-function, to A arrows, and to F forks. Observes that from the second requirement it follows that forks can connect only vertices that have the same time value (set by T).

Definition 6.2 (History, slice.)**.** For a time graph (G, T) and a vertex $u \in V$, let G_u be the subgraph of (V, A) induced by $\{v \in V : T(v) \leq T(u)\}$. We will denote by H_u the 'History of u', which is the set of vertices that are connected to u in G_u. A slice is an equivalence class of the relation $u \equiv v$ if and only if $H_u = H_v$. Notice that the time function has to be constant on a slice, to see this consider vertices x, y with $T(x) > T(y)$, thus $x \notin G_y$ and therefore $x \in H_x / H_y$ So they are belong to different classes. Thus, we can extend the time function and the history to slices and write for a slice K, $T(K)$ and H_K.

Claim 6.3. Consider slices K_1, K_2 such that $T(K_1) = T(K_2)$, then either $K_1 = K_2$ or H_{K_1}, H_{K_2} are disjoint.

Proof. Assume a non-empty intersection, so there exists $z \in H_{K_1} \cap H_{K_2}$ such that for any $x \in K_1$ and $y \in K_2$, there exist arrow-paths $z \to x$ and $z \to y$. Thus, for any $w \in H_{K_1}$, one can concatenate a path $w \to x \to z \to y$ and conclude that $y \in H_{K_2}$. □

Definition 6.4 (The graph of slice)**.** Let K be a slice. The graph of K, denoted by $G_K = (V_K, E_K)$, is the graph whose vertices are slices $R \subset H_K$ with $T(R) = T(K) - 1$. Two vertices are connected by an edge if and only if there is a fork whose ends belong to the slices associated with them. Namely, $S, R \in V_K$ are connected in G_K if and only if there is $\{s, r\} = f \in F$ such that $s \in S$ and $r \in R$.

The connection property is crucial, In our construction, there are not neutral forks, and we will have to find way to overcome it, one way is to take $\arg\min_{u \in \sigma_1, v \in \sigma_2} d(u, v)$ and connect the by fork. More thought should be invested here.

Claim 6.5. G_K is connected.

Proof. Let $R, S \in V_K$ and consider $r, s \in V$ such that $r \in R, s \in S$. Recall that $R, S \subset H_K$ and therefore $r, s \in H_K$, Thus, by definition of the 'History' there exists a path, passing trough arrows only, from r to s in H_K. Consider such a path that is minimal in the number of points x belong to it, with $T(x) = T(K)$. By induction on the number k of such points we prove that R and S are connected in G_K.

1. Base. $k = 0$, In this case, the path from r to s lies within $G_r (= G_s)$. Hence $H_r = H_s$, So $R = S$.

2. Induction step. There exists y on the path from r to s such that $T(y) = T(K)$. Let x be the point which precedes y on the path, and z the point that follows y. Since the values of T at the two ends of an arrow differ by 1, and since y gets the maximal T value in H_K it follows that:

   (a) $\{x, z\} \subset \mathrm{pre}(y) \Rightarrow \{x, z\} \in F$.

   (b) The slices of x and z, say K_x and K_z has time value $T(K_x) = T(K_z) = T(K) - 1$, Namely they are both members of V_K.

Since the path connecting r, x in H_K has $k-1$ vertices with $T(\cdot) = T(K)$, then by the induction hypothesis, R and K_x are connected in G_K, and by similar argument so are K_z and S. On the other hand, $\{x, z\} \Rightarrow \{K_x, K_z\} \in E_K$. Thus, R and S are connected in G_K.

$\square$

Definition 6.6. Let $f\colon G \hookrightarrow \mathbb{R}^d$ be an embedding that maps each element of $V \cup A \cup F$ to a point in $\mathbb{R}^d$. We say that $\left(G, \mathbb{R}^d\right)$ is a space-time graph if and only if:

1.
2. The point set by f for an edge is the mean of its endpoints' images; that is, for $e = \{x, y\} \in A \cup F$, we have $f(e) = \frac{1}{2}\left(f(x) + f(y)\right)$.

Definition 6.7. Define $\mathrm{pre}_i(v) = \arg\max_{u \in \mathrm{pre}(v)} L_i(f(u))$

Claim 6.8. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned}
\mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\
&\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\
&= \mathbf{Size}(R_1) + \mathbf{Size}(R_2)
\end{aligned}$$

$\square$

Definition 6.9 (Spanned Set)**.** Consider an embedding $f\colon \star \hookrightarrow \mathbb{R}^d$, a subset of f's domain $S \subset \star$, and $d+1$ points $x_1, \ldots, x_{d+1}$. We say that $\langle S, x_1, \ldots, x_{d+1}\rangle$ is a spanned set of S if for any $i \in [d+1]$ and $s \in S$, it holds that $L(f(s)) \leq L_i(x_i)$ and in addition there are $v_1, ..v_{d+1} \in V$, not necessary different, that achieve the equalities $L_i(x_i) = L_i(f(x_i))$. We name $x_1, \ldots, x_{d+1}$ the poles of S. Furthermore, we extend the notation to a spanned slice when S is a slice of some time-graph. Notice that saying $\langle S, x_1, \ldots, x_{d+1}\rangle$ is a spanned set means that $\mathbf{Size}(S) = \mathbf{Size}(\{x_1, \ldots, x_{d+1}\})$.

Observe that in the case where the subset S is a slice. Each of these poles can be associated with a vertex of G that gives the equality. If there are multiple vertices whose images by f give equality, associate the point with one of them arbitrarily. We will use the notion of applying functions originally defined over vertices to the poles, referring to the application over the vertices associated with them. For example $\mathrm{pre}(x_i) = \mathrm{pre}(v)$ for the vertex $v \in S$ which satisfies $L_i(f(v)) = L_i(x_i)$ and therefore is associated with the pole x_i. Another example is referring to x_i as a vertex, so when saying 'For $x_i \in V_k$' or 'For $\{x_i, u\} \in F$', we refer to the associated vertex or the edge connecting it to u.

Claim 6.10. Consider a time graph (G, T), an embedding $f\colon G \to \mathbb{R}^d$, such that.. , Let K be a slice in G, and let $\hat{K} = \langle K, x_1, ..x_{d+1}\rangle$ be a spanned slice of K such that $K \neq H_K$. Then for some integer k there exist spanned slices $\hat{K}_0 = \langle K_0, \cdot\rangle, .., \hat{K}_k = \langle K_k, \cdot\rangle$ and Forks $F_1, .., F_k$ such that:

1. For $i \in [d+1]$ $\text{pre}_i(x_i)$ belongs to the poles $\hat{K}_0, .., \hat{K}_k$.
2. Introduce an edge between $\hat{K}_i$ and $\hat{K}_j$ if and only if one of the Forks $F_1, .., F_k$ consists of a pole of $\hat{K}_i$ and a pole of $\hat{K}_j$. Then the graph formed from $\hat{K}_0, .., \hat{K}_k$ is connected.
3. $\sum_{i=0}^{k} \textbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \textbf{Size}(F_i) \geq \textbf{Size}(\hat{K}) + 1$

Proof. First Notice that K doesn't contain leaves in (V, A), since a leaf is connected only to itself via arrows, and therefore its 'slice' equals to its 'History'. Thus $\text{pre}_i(x_i)$ is not empty for $i = 1, .., d+1$. Next consider the graph $G_K = (V_K, E_K)$ from definition 6.4. Since $\{x_i, \text{pre}_i(x_i)\}$ is an arrow $\text{pre}_i(x_i)$ belongs to a slice in V_K. for $i = 1, .., d+1$ denote by R_i the slice in V_K that contains $\text{pre}_i(x_i)$.

By claim 6.5 G_K is connected and therefore there exist a minimal collection $\{K_0, .., K_k\}$ of slices form V_K which contains $R_1, R_2, ..R_{d+1}$, and which is connected in G_K. We pick a set of forks $\mathbf{F} = \{F_1, .., F_k\}$ which realizes a spanning tree for this collection of slices. Later we will identify edges from this spanning tree with the forks from $\mathbf{F}$.

For $i = 1, .., d+1$ we set $\text{pre}_i(v_i)$ to be the ith pole of R_i, This assures (1). The set of remaining poles for $\hat{K}_0, .., \hat{K}_k$ will be equal to $\cup_{i=k}^{k} F_i$. This will assure (ii), We will also select poles for $F_1, .., F_k$ to form 'spanned forks' $\hat{F}_1, \cdots, \hat{F}_k$ in such way that:

$$\sum_{i=0}^{k} \textbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \textbf{Size}(\hat{F}_i) \geq \sum_{i=1}^{d+1} L_i\left(\text{pre}_i(v_i)\right) \geq \textbf{Size}(\hat{K}) + 1$$

This will assure (3).

Let us denote by $J_0, .., J_{2k}$ the enumerated union $K_0, ..K_k, F_1, .., F_k$. Observes that for any choice of $(d+1)(2k+1)$ elements $z_0^1, z_0^2, z_0^3, .., z_0^{d+1} .., z_{2k}^{d+1}$ in $\mathbb{R}^d$, such that $z_j^i \in J_j$ it holds that:

$$\sum_{i=0}^{k} \textbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \textbf{Size}(\hat{F}_i) \geq \sum_{j,i} L_i(z_j^i) \tag{2}$$

Later, we are going to take the points $\{z_j^i\}$ from the intersection of J_j's. For convenience, let us assume that for the index subset $X \subset [2k] \cup \{0\}$, the intersection $\bigcap_{j \in X} J_j$ returns a single point. If there is more than one, then decide on the point in a way that all the chosen points simultaneously satisfy that for any index subsets Y, X, such that $X \cap Y$ and they both introduce a non-trivial intersection $\bigcap_{j \in Z} J_j$: $Z \in \{X, Y\}$, then:

$$\bigcap_{j \in X} J_j = \bigcap_{j \in Y} J_j$$

Now consider a non-trivial maximal intersection $\bigcap_{j\in X} J_j$, the intersection is maximal in the sense that for any $j' \notin X$ it holds that .$\bigcap_{j\in X\cup\{j'\}} J_j = \emptyset$. We claim that for any i and for any such X, there is $t \in X$ such that J_t is the successor of J_j on the path to R_i for any $j \in X/\{t\}$. Assume not. Let $x, y \in X$, and denote the paths from J_x, J_y to the ith pole by $\langle J_x, J_{x_2}, \dots, R_i\rangle$, $\langle J_y, J_{y_2}, \dots, R_i\rangle$, such that $J_y \neq J_{x_2}$. If $y = x_2$, $y_2 = x$, or $x_2 = y_2 \in X$, then pick other x, y (if there is no such, we get an immediate contradiction). In that case, $\langle J_x, J_y, J_{y_2}, \dots, R_i\rangle$ is a path from J_x to R_i which is different from $\langle J_x, J_{x_2}, \dots, R_i\rangle$. Therefore, there are at least two paths from J_x to the ith pole, namely, there is a cycle in contradiction to the minimality of $K_0, \dots, K_k$.

We will choose the z^i_j by the following process. Pick a z^i_j that has not been set yet. If $J_j = R_i$, then set $z^i_j \leftarrow \mathrm{pre}_i(x_i)$ - we call this a type-1 assignment. Otherwise, consider the path from J_j to R_i (by connectivity, it has to be such a one, and by minimality, there is exactly one). Let J_t be the successor on the path, then pick $z^i_j \in J_j \cap J_t$ - similarly, we call this assignment a type-2 assignment. Since any type-2 assignment is associated with a non-trivial intersection, and any non-trivial intersection induces a type-2 assignment for all i's (since for all i's, one of the J_j on its support is a successor of the rest on their path to the ith pole), we have that:

$$\begin{aligned}
\sum_{j,i} L_i(z^i_j) &= \sum_{j,i,z^i_j\in\text{type-1}} L_i(z^i_j) + \sum_{j,i,z^i_j\in\text{type-2}} L_i(z^i_j) \\
&= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(x_i)\right) + \sum_{X:\,\bigcap_{j\in X} J_j\neq\emptyset} (|X|-1) \sum_{i}^{d+1} L_i\left(\bigcap_{j\in X} J_j\right) \\
&= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(x_i)\right)
\end{aligned}$$

Plugging it into eq. (2) gives the desired. □

7 Explantation of a Spanned Slice

[**COMMENT**] Here we follow the original proof, but change the triminalogy, m will be the number of leaves in the explanation, then in the end we conclude that the number of faults is at least m/Δ. So $|R| \leq f(\frac{m}{\Delta})$.

Definition 7.1. Consider a time graph (G, T), an embedding $f\colon G \to \mathbb{R}^d$, such that.. Let $\beta = \frac{d+1}{\xi}$ and $\alpha = \beta\mathbf{Size}(\text{fork})$. Given sets W of points, R of arrows and F of forks, we define U as the set of vertices of the graph induced by the edges of $R \cup F$. The sets W, R and F form an explanation of a spanned slice $\hat{K} = \langle K, v_1, .., v_{d+1}\rangle$ if and only if:

1. $W \cup \{v_1, .., v_{d+1}\} \subset U \subset H_K$ and the graph $(U, R \cup F)$ is connected.

2. All elements of W are leaves.

3. For $m = |W|$ and $s = \mathbf{Size}(K)$ we have $|R| \le \alpha(m-1) - s$ and $|F| < m - 1$.

Claim 7.2. Every spanned slice $K = \langle K, v_1, .., v_{d+1} \rangle$ has an explanation.

Proof. Let $h = T(K) - \min_{u \in H_K} T(u)$. The proof is by induction on h.

1. Base: $h = 0$, namely T is constant on H_K, so no arrows connect points of H_K. Thus, H_K, as well as $\mathbf{K}$, consists only of leaves. In that case, $U = W = \{u\}$, $m = 1$, $s = 0$, and $|R|, |F| = 0$ satisfy the requirement.

2. Induction Step: $h \neq 0$ implies $K \neq H_K$. Thus, by claim 6.10, we know that for some k there exist spanned slices $\hat{K}_0, .., \hat{K}_k$ and forks $F_1, .., F_k$ that satisfy claim 6.10.

   Consider $i \in 0, .., k$, denote by $s_i = \mathbf{Size}(\hat{K}_i)$. By the inductive hypothesis and since $T(K_i) = T(K) - 1$, we know that $\hat{K}_i$ has an explanation formed by a set of leaves $\mathbf{W}_i$, a set of arrows $\mathbf{R}_i$ and a set of forks $\mathbf{F}_i$. Let $m_i = |\mathbf{W}_i|$, So the sets R_i and F_i have at most $\alpha m_i - s_i$ and $m_i - 1$ leaves. We define:

   (a) $\mathbf{W} = \cup_{i=0}^{k} \mathbf{W}_i$, Notice that $\mathbf{W}_i \subset K_i$ and that $T(K_i) = T(K_j)$ for all i, j, thus by claim 6.3 $\mathbf{W}$ is a union of disjoint set, Hence W contains $m = \sum_{i=0}^{k} m_i$ leaves.

   (b) $\mathbf{F} = \{F_1, .., F_k\} \cup_{i=0}^{k} \mathbf{F}_i$ with at most $k + \sum_{i=0}^{k} m_i - 1 = m - 1$ elements.

   (c) $\mathbf{R} = \{\{v_i, \text{pre}_i(v_i)\}\} \cup_{i=0}^{k} \mathbf{R}_i$ with at most

   $$\begin{aligned} &d + 1 + \sum_{i=0}^{k} \alpha(m_i - 1) - \beta s_i \\ \le& d + 1 + \alpha m - \alpha(k+1) - \beta\left(s + \xi - k \cdot \mathbf{Size}(\text{fork})\right) \\ =& \alpha\left(m - 1\right) - \beta s + \left(\alpha + d + 1 - (k+1)\alpha - \beta\xi + \beta k \cdot \mathbf{Size}(\text{fork})\right) \end{aligned}$$

   So its enough to show that for $k \ge 0$ the term in the closure is negative, We do that by showing that the slop is negative, and then the value of the term at zero is negative:

   $$-\alpha + \beta\mathbf{Size}(\text{fork}) \le 0 \Rightarrow \mathbf{Size}(\text{fork}) \le \frac{\alpha}{\beta}$$

   $$d + 1 - \beta\xi \le 0 \Rightarrow \frac{d+1}{\beta} \le \xi$$

   And those inequalities hold for $\beta = \frac{d+1}{\xi}$ and $\alpha = \beta\mathbf{Size}(\text{fork})$. Finally, by claim 6.10, the graphs induced by $\mathbf{R}_i \cup \mathbf{F}_i$ are connected, thus the graph induced by $\mathbf{R} \cup \mathbf{F}$ is also connected. Therefore, $\mathbf{W}, \mathbf{R}, \mathbf{F}$ form an explanation of K.

□

8 Counting Subtrees.

Claim 8.1. Let $G = (V, E)$ be a $\Delta + 1$-regular graph, and let k be an integer less than $|E|$. Fix a vertex $r \in V$. The number of subgraphs rooted at r with less than k edges is at most α^k.

Proof. Denote by T_Δ and T_2 the infinity Δ and binary tress. There is an injection between the subtrees of G, rooted at v, with at most k edges, to the subtrees of T_Δ rooted at v, with at most k edges. Moreover, there is an injection from the subtrees of T_Δ rooted at v, with at most k edges to the subtrees of T_2, rooted at v, with at most $k \log \Delta$ edges.

[COMMENT] In the second map, we think on replacing each of the vertex with $\Delta - 1$-leaves binary tree, each leaf is associated with a outgoing edge. .

Thus, it's enough to bound the number of of subtrees in T_2 rooted at v, with at most $k \log \Delta$ edges, that number is less than the number of binary tress with $k \log \Delta + 1$ leaves. For exact leaves number we get the $k \log \Delta$ Catalan number, which its limit is:

$$\sim \frac{4^{k \log \Delta}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}}$$

Therefore we can bound by:

$$\sum_{j=1}^{k} \frac{\Delta^{2j}}{(j \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta} \sum_{j=1}^{k} \frac{1}{j^{\frac{3}{2}}} \leq \zeta(\frac{3}{2}) \cdot \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$

$\square$

Now, from here we get an upper bound on the number of subgraphs that connected l vertices at the top level. We bound it from above by counting combination of rooted subtrees, such the vertices are their root and the overall edges number is summed up to k. Thus:

$$\leq \zeta(\frac{3}{2}) \binom{k-1}{l-1} \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$

9 Tree Separator Lemma

For a tree $T = (V, E)$ and a vertex $v \in V$, we call the subtree obtained by taking the descendants of v the subtree rooted at v.

Claim 9.1. Let $T = (V, E)$ be a tree with $m > 3$ leaves and τ edges, then it has a subtree with $m' \in (\frac{1}{3}m, \frac{2}{3}m)$ leaves and at most $\frac{m'}{m}\tau$ edges.

Proof. Order the vertices according to their heights. Denote by $v \in V$ the lowest vertex for which the subtree rooted at v has at least $\frac{1}{3}m$ leaves. Now, denote by T_1 the subtree obtained through the following iterative process over v's children:

1. Initialize T_1 as the subtree rooted at v.

2. For a child u of v:

   (a) Check if erasing from T_1 the subtree rooted at u doesn't decrease the number of leaves below $\frac{1}{3}m$. If so, then we update T_1 to be the result of the erasure.

Clearly, the algorithm is defined such that T_1 has to have at least $\frac{m}{3}$ leaves. Furthermore, since any child of v is a root of a subtree with fewer than $\frac{m}{3}$ leaves (otherwise we get a contradiction to the minimality of v's height), we have that if T_1 would have more than $\frac{2}{3}m$ leaves, then the subtraction of any of its children would yield a tree with at least $\frac{2}{3}m - \frac{1}{3}m = \frac{1}{3}m$ leaves. Namely, v has in T_1 a child u such that the subtraction of the subtree rooted at u from T_1 results in a subtree with more than $\frac{1}{3}m$ leaves. But if indeed there were such a child, then the algorithm would erase it, so there is not.

Now let $T_2 = T/T_1 \cup v$, namly the tree induced by the subtraction of T_1's edges from T. Observes that since the summation of leaves in T_1 and T_2 is the leaves in T, and T_1 has no more than $\frac{2}{3}m$ leaves, and no less than $\frac{1}{3}$ leaves, it holds that the number of leaves in T_2 is also in the range $(\frac{1}{3}m, \frac{2}{3}m)$. We claim that at least one of the subtrees has fewer than $\frac{m'}{m}\tau$ edges.

To see this, denote by m_1, m_2 and by τ_1, τ_2 the leaves and the edges in T_1, T_2 respectively. Now:

$$\frac{\tau}{m} = \frac{\tau_1 + \tau_2}{m_1 + m_2} = \frac{m_1}{m_1 + m_2}\frac{\tau_1}{m_1} + \frac{m_2}{m_1 + m_2}\frac{\tau_2}{m_2}$$

Namely, $\frac{\tau}{m}$ is a weighted average of $\frac{\tau_1}{m_1}$ and $\frac{\tau_2}{m_2}$, and therefore it is greater than their minimum. Peak the subtree with the minimal quotient. $\square$

By repeating recursively on the above construction, we get the following corollary:

Corollary 9.2. Let $T = (V, E)$ be a tree with m leaves and τ edges. Then, for any $\alpha > 0$, it has a subtree with $m' \in \left(\frac{1}{3}\alpha m, \frac{2}{3}\alpha m\right)$ leaves and at most $\frac{m'}{m}\tau$ edges.

Claim 9.3. Let $\gamma, \beta \in \mathbb{R}_{>0}$ and n be a positive integer. For any tree T with $\tau \geq \beta n$ edges and m leaves such that $\tau \leq \gamma m$, there is a subtree T' of T with a number of edges τ' less than $\frac{\beta}{2}n$ and a number of leaves greater than $\frac{\beta}{4\gamma}n$.

Proof. Since the construction of claim 9.1 preserves the ratio of edges to leaves, we have that for a T' obtained using the construction, the number of edges satisfies $\tau' \leq \gamma m'$. So, it's enough to look for the maximal α such that:

$$\gamma m' \leq \gamma \frac{2}{3}\alpha m \leq \frac{\beta}{2}n \Rightarrow \alpha = \frac{3\beta}{4\gamma}\frac{n}{m}$$

Which gives $m' \geq \frac{1}{3}\alpha m = \frac{\beta}{4\gamma}n$.

$\square$

Move this definition to the beginning of the paper.

Definition 9.4. We model the qubits on the finite toric with a side length n by the projection:

$$(x_1, x_2, .., x_d) \mapsto (x_1 \bmod n, x_2 \bmod n, .., x_d \bmod n)$$

Rewrite when we talk about the first time we have deviation clusters of size $\in (\alpha d, d)$. The reason is that having those assumptions means all the droplets in the past shrank; thus, we can use the ratio of edges to time-leaves in the big subgraph.

Claim 9.5. Assume that until step τ, there was no deviation cluster of size greater than d. Let $\hat{K}$ be a spanned slice induced by a deviation of a cluster state at step τ, and let ω be an explanation for $\hat{K}$ such that the number of vertices in ω is greater than βn. Then there is a subgraph (subtree) ω' of ω such that ω' has fewer than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ leaves.

The connection between the size of the clusters and whether they shrink or not has not been written yet. Moreover, it seems that there might be confusion since two disjoint clusters might be time-connected. It seems that might be a problem.

Proof. Let ω be an explanation of $\hat{K}$ with more than βn vertices. Since there is no deviation cluster of size greater than d until step τ, we have that every vertex of ω is a spanned slice induced by a boundary of a deviation cluster of size at most d. Thus, ω satisfies the shrink property in Claim 15.6, and by Claim 7.2, the ratio of edges to leaves of ω is at most γ. Observe that a spanning tree of ω has a ratio of edges to leaves that is at most ω's ratio, and the number of edges equals ω's vertices minus 1. Therefore, by applying Claim 9.3 on ω's spanning tree, ω has a subtree with less than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ leaves. $\square$

State a claim, and notice that in the new version we are not only interested in trees whose root is a single vertex.

Proof. Now, since any two connected vertices are at a space-distance of at most 1, it follows that all the leaves in the subtree are at a space-distance of at most βn from each other. Thus, if $\beta < 1$, no two of them have the same space-coordinate modulo n. And therefore the probability to have such subtree is at most $p^{\frac{\beta}{4\gamma}n}$.

Because any ω with more than βn vertices implies the existence of a subtree with size less than $\frac{\beta}{2}n$, it's enough to bound the probability of having such a subtree. Considering that the trees are invariant to shifting by $n \cdot \mathbf{1}_i$, we have that it's sufficient to show that there are no such trees which are rooted in the finite cube $\mathbb{Z}_n^d \times \mathbb{Z}_t$. Finally, we use claim 8.1 to bound the number of such trees given a fixed root. Overall, we get:

$$\mathbf{Pr}\left[\text{No } \omega\text{s with more than } \beta n \text{ vertices}\right] \le n^d t \left(\xi^2 p^{\frac{\beta}{4\gamma}}\right)^n$$

$\square$

10 Tanner - History Connected Equivalency

Imagine the following process, we apply t noisy iterations of the SWEEP rule, and then a single ideal (without noise) iteration. Denote by ξ a deviated configuration (can be either bits or checks), and by ξ' the result of applying an ideal SWEEP rule on ξ. Notice that if ξ is connected in the Tanner graph then ξ' is connected in the History graph **[COMMENT]** Require proof, For the classical case enumerate all the options can be done by hand. Thus:

$$\begin{aligned}
&\mathbf{Pr}\left[\xi \text{ connected in Tanner}\right] \leq \mathbf{Pr}\left[\xi' \text{ connected in Time} \mid \text{no noise at the } t+1 \text{ iteration}\right] \\
&\leq \sum_{m}^{\infty} |\{\text{ explanation to } \xi' \text{ over } m \text{ leaves restricted to no noise at the final iteration }\}| p^{\frac{m}{\Delta}} \\
&\leq \sum_{m}^{\infty} |\{\text{ explanation to } \xi' \text{ over } m \text{ leaves }\}| p^{\frac{m}{\Delta}} \\
&\leq \mathbf{Pr}\left[\xi' \text{ connected in Time } \textbf{- Upper Bound }\right] \leq q^{\mathbf{Span}(\xi')} \leq q^{\mathbf{Span}(\xi)-1}
\end{aligned}$$

10.1 Another Idea.

Reconsider the construction of the explanation. We could add at the top an imaginary layer in which all the vertices at time t are connected by arrows to all the vertices (at time $t+1$) that are adjacent to them in the Tanner graph. For the last layer, we might find that the span of a slice expands instead of shrinking. Namely, claim 6.10 for the final slices would change to:

$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \mathbf{Size}(\hat{K}) - \xi$$

Thus, following the computation at the induction stage in claim 7.2, the number of arrows is bounded by:

$$\begin{aligned}
&\alpha\left(m-1\right) - \beta s + d + 1 + \alpha + \beta\xi \\
&\alpha\left(m-1\right) - \beta s + O(1)
\end{aligned}$$

11 Intermediate Step Lemma

[COMMENT] The idea of this step is also to bound the case that all the bits are flipped at once, So checks are satisfied (and the argue above is not applied directly).

Definition 11.1 (Connected Cluster With Span s). Let $G = (B, C, E)$ be a Tanner graph of a code, and let $f \colon G \to \mathbb{R}^{d+1}$ be an embedding of the graph into $\mathbb{R}^{d+1}$. For $s \in \mathbb{R}$, we say that $X \subset B$ is a connected cluster of G with span s regarding f if X is connected in G (where two bits are connected if there is a check that connects both of them in E) and $f(X) \subset \mathbb{R}^{d+1}$ has $\mathbf{Size}(f(X)) \leq s$. When it's clear from the context which G and f are, we will call X in brief a connected cluster with span s.

Definition 11.2. The model, computation, and noise are tossed, and then a correction is made.

Claim 11.3. Let ξ be the event in which, at the **beginning** of time t, the error contains a connected cluster with a span greater than $\frac{d}{4}$, and for $i \in [t]$, let σ_i be the event in which, at the **end** of time i, there is a connected cluster with a span of at least $\sqrt{d}$ that belongs to the error. Then:

$$\mathbf{Pr}\left[\xi\right] \leq t\mathbf{Pr}\left[\sigma_1\right] + p^{\frac{1}{8}\sqrt{d}}$$

Proof. By the law of total probability it hodls:

$$\mathbf{Pr}\left[\xi\right] = \mathbf{Pr}\left[\xi \cap \bigcup_i^t \sigma_i\right] + \mathbf{Pr}\left[\xi \cap \bigcap_i^t \bar{\sigma}_i\right]$$

First, let us bound from above the probability of the right intersection, In that case, at the **end** of time $t-1$, the erorr consists of clusters each at span less than $\sqrt{d}$. Let m be the number the clusters and denote them by $X_1, .., X_m$. Now since the clusters are disjoint, the sum of their $\mathbf{Size}(X_i)$ is lower than the total number of the bits, namely:

$$\sum_1^m \mathbf{Size}(X_i) \leq n \Rightarrow m \leq \frac{n}{\sqrt{d}}$$
$$p\frac{n}{\sqrt{d}} \leq \sqrt{d}$$

, Thus, . So denote by τ the number of bits that require to be flipped in order to merge the flipped connected component to a one with a span greater than $d/4$, So τ has to be at least:

$$\tau\left(\sqrt{d}+1\right) \geq \frac{1}{4}d \Rightarrow \tau \geq \frac{1}{8}\sqrt{d}$$

Therefore:

$$\begin{aligned}\mathbf{Pr}\left[\xi \cap \cap_i^t \bar{\sigma}_i\right] &\leq \sum_{\xi^{(t-1)}} \mathbf{Pr}\left[\xi | \xi^{(t-1)}\right] \mathbf{Pr}\left[\xi^{(t-1)} \cap \cap_i^t \bar{\sigma}_i\right] \\ &\leq \sum_{\xi^{(t-1)}} p^{\frac{1}{8}\sqrt{d}} \mathbf{Pr}\left[\xi^{(t-1)} \cap \cap_i^t \bar{\sigma}_i\right] \leq p^{\frac{1}{8}\sqrt{d}}\end{aligned}$$

$$\begin{aligned}\mathbf{Pr}\left[\xi\right] &= \mathbf{Pr}\left[\xi \cap \cup_i^t \sigma_i\right] + \mathbf{Pr}\left[\xi \cap \cap_i^t \bar{\sigma}_i\right] \\ &\leq t\mathbf{Pr}\left[\sigma_1\right] + p^{\frac{1}{8}\sqrt{d}}\end{aligned}$$

□

Definition 11.4. The atomic events are the sets of space-time locations in which faults occur. A run ω is the sequence of states (assignment of all the bits) induced by an atomic event (and the transition function of the machine). We say that a run ω closes a non-trivial cycle if it contains a state in which there is a connected cluster A such that its boundary is not a co-boundary.

Define the bits-cluster graphs $G = (V, E)$ such that the vertices are connected bit-clusters at different times $\{A_i^t\}$, and there is an edge between A_i^t and A_j^{t-1} only if $\Pi_x A_i^t \cap \Pi_x A_j^{t-1} \neq \emptyset$. Given a state at time $t-1$ such that no connected component in G forms a closed non-trival circle, consider $\tilde{A}_i^t$ constructed as follows: perform the application of either noise or Toom's correction in an arbitrary order and stop right before $\tilde{A}_i^t$ forms a closed non-trivial circle.

Claim 11.5. There is $\phi \in Aut(\mathbb{T}_n^d)$ for which $\mathbf{Size}(\phi(\tilde{A}_i^t)) \geq n-1$.

Claim 11.6. Consider a running in which no non-trivial circle was closed, then there is $\psi \colon \mathbb{T}_n^d \to \mathbb{Z}_{2n}^d$ such:

1. For any slice K, $\psi(K)$ is connected in $\mathbb{Z}_{2n}^d$.
2. For any t, denote by S_t the state at time t. Then, for any $t > 1$, we have that $\psi(S_t)$ is the image of the application of the Tooms rule over $\psi(S_{t-1})$.

Proof. We prove it by induction on time t.

1. Base case, $t = 1$. We will construct ψ in the following iterative manner. First, let ψ be the identity. Then, while there are slices that are not connected on $\mathbb{Z}_{2n}^d$, do the following: Let K be a slice that is not connected on $\mathbb{Z}_{2n}^d$. Then it can be decomposed into a disjoint union $K = K_1 \cup K_2$ such that K_1 and K_2 are connected on $\mathbb{Z}_{2n}^d$. Assume, without loss of generality, that for a coordinate $x \in [d]$, it holds that for any $a \in K_1$ and $b \in K_2$, we have $b_x - a_x > 1$.

   Observe that there has to be at least one such coordinate; otherwise, it is a contradiction to K_1 and K_2 not being connected in $\mathbb{Z}_{2n}^d$. Then define:

   $$\psi(z) \leftarrow \begin{cases} \psi(z) & z \cap K_1 = \emptyset \\ z + n\mathbf{1}_x & \text{else} \end{cases}$$

   Repeat the above for other coordinates if $\psi(K)$ is still not connected in $\mathbb{Z}_{2n}^d$. The proof that $\psi(K)$ has to be connected (that the process ends) is omitted.

2. Induction step. We repeat the construction in the base case. To prove the second property, we show that ψ maps (un)satisfied edges to (un)satisfied edges. **[COMMENT]** That is wrong.

□

Claim 11.7. Denote by ω running of the above type, then there is a running ω' on $\mathbb{Z}_\infty^d$ such that ω and ω' agree.

Claim 11.8. Consider a t-steps running, conditioned on the event that no trivial-cycle was closed in the $t-1$ first steps, the probability that a non-trivial cycle will be closed at the tth step is less than:

$$\mathbf{Pr}\left[\Lambda^t | \bigcap_{t'<t} \bar{\Lambda^{t'}}\right] < n^{2d} \sum_{i=n-1}^{\infty} p^i \mathcal{J}_i \leq n^{2d} q^{n-1}$$

Proof. Consider the event in which there is a droplet that at time t closes a non-trivial cycle. Let $\tilde{A}$ be the subset of that droplet obtained by the process in claim 11.5. By the claim, $\tilde{A}$ has a boundary σ and two vertices on that boundary $M = \{u, v\}$ such that $\mathbf{Size}(M) > n-1$, up to isomorphism.

Now, since no slice in the investigation of M closes a non-trivial cycle then by claim 11.6 .. so by claim 11.7 .. □

12 Configuration up to $C_z^{\perp}$

13 Drafts:

In addition since, in a tree, the number of leavs is at least greater than the maximal degree, then we have that each J_j intersects with at most $d+1$ elements.

$$\begin{aligned} L + \Delta_{\max} + 2(n-L-1) &\leq \sum_{v \in V} \deg(v) = 2(n-1) \\ \Rightarrow L &\geq \Delta_{max} \end{aligned}$$

14 Decoder Phase

In the general picture, I would like to use the 4D toric code to encode in any target code. We use the original fault tolerance construction (the concatenation construction) to encode into and decode from the 4D toric code. So, for the decoding phase, we need to have a decoder which, in the ideal setting, runs in less than logarithmic time.

What we know so far:

1. We can assume that two unsatisfied checks are belong to the same history with probability less than $\sim q^{|u-v|}$.

2. There is a $\log^2(n)$ depth algorithm to find the perfect/max matching in a general graph. From first check, the known algorithms to deal with the weight matching are based on LP.

3. Applying Toom's/Sweep rule will require at least linear depth.

4. Having an algorithm that succeeds with constant probability $\frac{1}{2} + \epsilon$ might be enough (Anyway the qubit at the result of the decoding is going to absorb p-noise).

15 Appendix

Definition 15.1. Let z be a connected assignment, and let Γ be the set of vertices adjacent to unsatisfied checks induced by z. For a generator $g \in S$, we say that v is a g-pole if $v \in \Gamma$ and $gv \notin \Gamma$. Furthermore, if there is a vertex $v \in \Gamma$ such that for any $g \in S$, we have $g^{-1}v \notin \Gamma$, then we say that v is a local minimum of z.

Let $x_1, .., x_k \subset \{\pm\}^k$ and let $p = \prod_i^k g_i^{x_i} v$ we say that p is a forward path in z if it holds that:

1. Any partial product of the first j elements $\prod_i^j g_i^{x_i} v$ is a vertex in z.
2. For any $g \in S$, we have that the sign of the sum of the exponents of the powers of the elements in the product associated with g is positive. Namely:

$$\sum_{i: g_i = g} x_i > 0$$

Similarly, we say that p is a backward path if the above holds, but the sign in the second condition is either negative or zero. The infimums of z are defined to be the set of vertices $U \subset z$ such that for any $u \in U$ and $v \in z$, other vertices in z, there is a forward path starting at u and ending at v.

15.1 Alternative Proof For $d' = 2$.

Definition 15.2. We denote by ϕ^t the subsets of bits which are flipped as a result of applying the Toom's rule at time t, and by ϕ_v the bits which are flipped by vertex v. We say that u is added to Γ^{t+1} by v at time t, if $u \in \Gamma^{t+1}$ and $u \in V(\phi_v)$. When it clear from the context, we will omit the time, and say that u is added by v.

Claim 15.3. Let $v \in \Gamma^t$ for any u that is added by v, and for any $x \in [d+1]$, we have $L_x(u) \leq L_x(v) + 1$.

Proof. By definition, any u that is added by v is of the form $v + \sum \alpha_j 1_j$, where $\alpha_j \in \{0, 1\}$ and $\sum \alpha_j \leq 2$. Thus, for any $x \in [d]$:

$$L_x(u) = \langle 1_x, v + \sum \alpha_j 1_j \rangle = \langle 1_x, v \rangle + \alpha_x \leq L_x(v) + 1$$

For $x = d + 1$ we have:

$$L_x(u) = \langle -\mathbf{1}, v + \sum \alpha_j 1_j \rangle = \langle -\mathbf{1}, v \rangle - \sum_j \alpha_j \leq L_x(v)$$

$\square$

Claim 15.4. Let $v \in \Gamma^t$ such that $L_{d+1}(v)$ is a (local) maxima in Γ^t then for any u which is added by v we have $L_{d+1}(u) \leq L_{d+1}(v) - 1$.

Proof. We will prove that $v \notin \Gamma^{t+1}$, and then by repeating on claim 15.3 where $\sum \alpha_j \geq 1$, we get the desired result. Assume, toward contradiction, that $v \in \Gamma^{t+1}$, and denote by w the vertex that adds v. If $w \neq v$, then there is at least one coordinate i for which $w_i < v_i$, thus $L_{d+1}(w) > L_{d+1}(v)$, in contradiction to $L_{d+1}(v)$ being a maximum in Γ^t. In the other case, where v adds itself, it follows that σ_v can't be decomposed into pairs of unsatisfied checks; namely, $|\sigma_v|$ is odd. We show by induction that impossible,namely that $|\sigma_v|$ is always even. The induction is over the size of the plaquettes subset that induce the boundary σ.

1. Base. Consider a boundary that induced by a single plaquette, in that case it obvious that $|\sigma_v|$ is either zero (where v is not belong to the plaquette) or 2.

2. Assume for any boundary that induced by at most k plaquettes, and consider a boundary σ which is induced by $k+1$ plaquettes, name this subset F and observes that F can be decomposed to a sum of two disjoint subsets A and B, each with less than $k+1$ plaquettes, Now:

$$\begin{aligned} \sigma_v &= (\partial A)_v \cup (\partial B)_v \\ \Rightarrow |\sigma_v| &= |(\partial A)_v| + |(\partial B)_v| - 2|(\partial A)_v \cap (\partial B)_v| = \text{even} \end{aligned}$$

   Where we used the induction assumption, which tells that $|(\partial A)_v|$ and $|(\partial B)_v|$ are even numbers.

□

Claim 15.5. Let $x \in [d]$ and let $v \in \Gamma^t$ such that $L_x(v)$ is a (local) maxima in Γ^t then for any u which is added by v we have $L_x(u) \leq L_x(v)$.

Proof. Since $L_x(v)$ is maximal, v doesn't see edges that goes from it at the positive x-direction, otherwise denote it by e and denote by w the second vertex at the end of e, w has an x-coordinate which is greater than v's x-coordinate by 1 and in addition $w \in \Gamma^t$ (since it lays on the support of e). Therefore $L_x(w) > L_x(v)$ in contradiction to $L_x(v)$ be maximal in Γ^t. Hence, any plaquette that is being flipped by v lies on edges associated with 1_j such that $j \neq x$. Thus, any vertex u that is added by v has the form $v + \sum_{j \in [d] \setminus \{x\}} \alpha_j 1_j$, where $\alpha_j \in \{0,1\}$ and $\sum \alpha_j \leq 2$. So:

$$L_x(u) = \langle 1_x, v + \sum_{j \in [d]/x} \alpha_j 1_j \rangle = \langle 1_x, v \rangle = L_x(v)$$

□

Claim 15.6. Let Γ^t be the vertices in the support of the domain wall σ^t. Denote $v_i = \arg\max_{v \in \Gamma^t} L_i(v)$. Then for any $i \in [d]$, one can find $u \in \Gamma^{t-1}$ such that u adds v_i and $L_i(u) \geq L_i(v_i)$. In addition, one can find $u_{d+1} \in \Gamma^{t-1}$ such that u_{d+1} adds v_{d+1} and $L_{d+1}(u_{d+1}) = L_{d+1}(v_{d+1}) + 1$.

Proof. Just take u_i to be any vertex that adds v_i. By claims claim 15.3, claim 15.4, and claim 15.5, we get the claim. □

[COMMENT] $V(\phi_v)$ was not defined. $v \leq w$ also not defined.

Definition 15.7 (σ_1 Precedes σ_0)**.** We say that σ_1 precedes σ_0 if, for any $v \in \sigma_1$, it holds that $\phi(v) \subset \sigma_0$.

Definition 15.8 (Explanation.)**.** Let σ_0 be a boundary at time t, and let $\sigma_1, \dots, \sigma_k$ be boundaries at time $t-1$, such that for $i \in [k]$, it holds that σ_i precedes σ_0. Then the explanation for σ_0 is the subgraph induced by the vertices of $\sigma_0, \sigma_1, \dots, \sigma_k$, the edges $\{u, v\} \in \sigma_i$, and the arrows $\{v, u\}$ for $v \in \cup\sigma_i$ and $u \in \phi(v)$.

The number of edges above is not lineary bounded by the leaves. Yet it also seems not minimal. So what we would like to do is to keep the poles, then to connect the islands by fork. So what is it a fork?

Definition 15.9 (Fork)**.** Let u, v be such that $u \in \sigma_1$ and $v \in \sigma_2 \neq \sigma_1$. If there exist $w \in \phi(v)$ and $z \in \phi(u)$ such that $\{w, z\} \in \mathcal{F}_1$, then there is a fork between v and u.

Proof. Idea: Suppose not, then $\sigma_0 = \phi(\sigma_1) \cup \phi(\sigma_2)$ is not connected. □

Claim 15.10. Assume that $\sigma_1, \sigma_2 \to \sigma_0$, then there is a fork between σ_1 and σ_2. So if we keep to extent the history in a recursive manner, then we get a connected graph.